UNIT-III
LASER AND FIBRE OPTICS
TWO MARKS

1. What is meant by population inversion? (April 2011, May 2012 2013 & 2015 Nov 2017,May 2018)

The process of achieving large population of atoms in the upper level (N_2>N_1) than the lower level is said to be population inversion.

2. What is called stimulated emission? (Jan 2013)

An atom in the ground state with energy E_1 absorbs a photon of energy hν and go to the exited state with energy E_2 provided that the photon energy hν is equal to the energy difference (E_1-E_2).This process is called stimulated emission.

3. Define spontaneous emission and stimulated emission. (Jan 2014, May 2014)

(i) Spontaneous emission:

The atom is the excited state returns to the ground state by emitting a photon of energy $\mathbf{E= E_2-E_1=h\upsilon}$**,** spontaneously without any external triggering. This process is known as spontaneous emission.

(ii) Simulated emission:

The atom in the excited state can also return to the ground state by external triggering (or) inducement of photon thereby emitting a photon of energy equal to the energy of the incident photon, known as stimulated emission.

4. Explain the classification of optical fibres based on the materials? (Jan 2013)

Based on material: glass fibre, plastic fibre

5. What are the characteristics features of a laser beam? (Jan 2010, May 2014)

The characteristics features of a laser beam are perfect monochromatic, extremely intense, High Directionality, Highly coherent.

6. Define total internal reflection? (May 2013, 2014, Jan 2016)

The phenomenon of total internal reflection takes place when it satisfies the following two conditions.

Condition 1: Light should travel from denser medium to rarer medium

i.e. $n_1>n_2$

Where n_1= refractive index of core
n_2= refractive index of cladding

Condition 2: The angle of incidence on core should be greater than the critical angle

i.e $\phi>\phi_c$

Where ϕ- angle of incidence
ϕ_c- Critical angle

7. Explain temporal coherence. (May 2011)

The maximum length of the resulting wave train on which any two points can be correlated is called temporal coherence.

8. What is the basic principle of fibre optics? (May2011,15, 17, Jan 2015,16)

Total internal reflection is the principle behind the transmission of light waves in an optical fibre.

9. Difference between spontaneous and stimulated emission. (May2014,17Jan 2016,Nov2016)

S.No	Spontaneous Emission	Stimulated Emission
1.	Emitted Photon travels in random direction	Emitted Photon can be made to travel in same direction.
2.	Emitted Photon cannot be controlled.	Emitted Photon can be controlled.

3.	Emitted Photon is less intense and is incoherent.	Emitted Photon is highly intense and is coherent.
4.	Emission of light radiation is not triggered by any external influence	Forced emission of light radiation caused by incident photons.

10. What are the different types of pumping in Lasers? (Jan 2015, May 2017)

Different Types of pumping are

- Optical pumping,
- Direct electron pumping,
- Inelastic atom-atom collisions and
- Chemical reaction.

11. State the principle of fibre optic communication. (May2014)

The basic principle of optic fibre communication is transmission of information over a required distance by the propagation of optical signal through optical fibre.

12. What is meant by pumping of atoms and mention types? (May 2014)

The excitation of atoms in order to achieve the population inversion can be done by the help of external energy is referred as pumping. Types are

Optical pumping, direct electron pumping, Inelastic atom-atom collisions, Chemical reaction.

13. What are the different types of laser? (Jan 2014)

Based on the active medium, they are classified as Solid state lasers, Gas lasers, Liquid Lasers, Dye lasers and Semiconductor lasers.

14. Mention any two applications of laser. (Jan 2014)

Laser beams are used to obtain a three dimensional lens less holograms.

Laser is used in printers for computer printouts.

15. What are the uses of optical resonator in Laser? (Jan 2016, May2016)

The optical resonator, or optical cavity, is two parallel mirrors with optical coating placed around the active medium which provides feedback of the light for amplification. Typically one will be a high reflector, and the other will be a partial reflector. The partial reflector is called the output coupler, because it allows some of the light to leave the cavity to produce the laser's output beam.

16. Give the classification of optical fibres based on material, mode and refractive index. (Jan 2014, 2016)

Based on modes: single mode fibre, multimode fibre

Based on material: glass fibre, plastic fibre

Based on refractive index: step-index fibre, graded-index fibre.

17. Define acceptance angle. (Jan 2015, Nov 2017)

The rays which are incident on fibre core within conical half angle φ_2 will be refracted into fibre core and hence propagate into the core by total internal reflection. This angle is referred to as acceptance angle.

18. What is refractive index profile? (May 2016)

A refractive index profile is the distribution of refractive indices of materials within an optical fibre. Optical fibre has a graded index profile in which the refractive index varies gradually as a function of radial distance from the fibre center.

19. Define numerical aperture? (May 2015, Nov 2017, Sept 2019)

It is defined as sine of acceptance angle.

$$NA = \sin i \quad (or) \quad NA = \sqrt{n_1^2 - n_2^2}$$

20. Is two level laser system possible? Justify? (Nov 2019)

It is not possible to obtain a population inversion with optical pumping because the system can absorb pump light that is gain energy only as long as population inversion, and thus light amplification is not achieved.

21. What are the three basic requirements for the laser system. (Sept 2020)

1 .Active medium
2. Pumping source
3. Optical resonator

PROBLEMS :

1. What is the ratio of the stimulated emission to spontaneous emission at a Temperature of 250°C for the sodium D line? (Given data λ = 5.9X10^{-7}m, h= 6.626X10^{-34}Js) (April 2011, Jan2015)

T=250°C=250+273=523

$$\frac{\text{Stimulated emission}}{\text{spontaneous emission}} = \frac{R_{21}\,(ST)}{R_{21}\,(SP)} = \frac{1}{e^{\left(\frac{h\nu}{K_B T}\right)} - 1}$$

$$\frac{\text{Stimulated emission}}{\text{spontaneous emission}} = \frac{R_{21}\,(ST)}{R_{21}\,(SP)} = \frac{1}{e^{\left(\frac{hc}{K_B \lambda T}\right)} - 1}$$

$$\frac{\text{Stimulated emission}}{\text{spontaneous emission}} = \frac{R_{21}\,(ST)}{R_{21}\,(SP)} = \frac{1}{e^{\left(\frac{6.625\times10^{-34}\times3\times10^{8}}{1.38\times10^{-23}\times523\times5890\times10^{-10}}\right)} - 1}$$

$$\frac{\text{Stimulated emission}}{\text{spontaneous emission}} = \frac{R_{21}\,(ST)}{R_{21}\,(SP)} = 5.329\text{x } 10^{-21}$$

2. Calculate the angle of acceptance and critical angle of an optical fibre whose refractive indices of the core and the cladding are 1.563 and 1.49 respectively (Jan 2016).

Solution: Refractive index of core is $n_1 = 1.563$
Refractive index of cladding is $n_2 = 1.49$

(i) Numerical aperture (NA) = $\sqrt{n_1^2 - n_2^2}$ =√(1.563^2-1.49^2)= 0.4721
(i) Acceptance angle= sin^{-1}(NA)= sin^{-1}(0.4721)= 28°53'
(ii) Critical angle ϕ_c=sin^{-1}[n_2/n_1]= sin^{-1}[1.49/1.563]= 72°41'

3. Calculate the wavelength of emission from GaAs whose band gap is 1.44eV. (May 2015, Jan 2016).

Solution: Bandgap of given GaAs semiconductor (E_g)= 1.44eV

= 1.44X1.609X$^{10-19}$

= 2.3X$^{10-19}$J

4. What is the ratio of the stimulated emission to spontaneous emission at a Temperature of 250°C for the sodium D line? (Given data λ = 5.9X10^{-7}m, h= 6.626X10^{-34}Js) (April 2011,Jamn 2015)

T=250°C=250+273=523

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$$\frac{\text{Stimulated emission}}{\text{spontaneous emission}} = \frac{R_{21}\,(ST)}{R_{21}\,(SP)} = 5.329\text{x } 10^{-21}$$

w.k.t., λ=hc/ E_g = $6.626X10^{-34}$ X $3X10^{8}/2.3X10^{-19}$ = $8.6X10^{-7}$m = 0.86 µm

5. Calculator the numerical aperture, acceptance angle and the critical angle of a fibre having core refractive index= 1.54 and cladding refractive index =1.50. (April 2018)

Given

Refractive index of core (n_1) =1.54

Refractive index of cladding (n_2) =1.50

Solution

Numerical aperture (NA) = $\sqrt{n_1^2 - n_2^2}$

$$NA = \sqrt{1.54^2 - 1.50^2}$$

NA = 0.347

Acceptance angle (i_m) = $\sin^{-1}$(NA)

(i_m) = $\sin^{-1}$(0.347)

(i_m) = 20..30 ^{o}

Critical angle $\phi_c = \sin^{-1}[n_2/n_1] = \sin^{-1}[1.50/1.54] = 76.91\,^o$

Part B Pondicherry University 11 marks Question and Answer

1. **Explain the principle and working of a semiconductor laser.(Jan 2016,May 2015,2016)**

Characteristics:

1. Type: Homo junction semiconductor laser
2. Active medium: P-N junction diode.
3. Active Centre: Recombination of electrons and holes.
4. Pumping: Direct pumping.
5. Optical Resonator: Junction of diodes- polished.
6. Power output: 1 mW
7. Nature of output: Pulsed (or) continuous.
8. Wavelength of the output: 8400Å- 8600Å.
9. Band gap: 1.44eV.

Principle:

The electron in conduction band combines with a hole in the valence band and hence the combination of electron and hole produces energy in the form of light. This photon in turn may induce another electron in the conduction band (CB) to valence band (VB) and thereby stimulate the emission of another photon.

Construction:

The active medium is a p-n junction diode made from a single crystalline material i.e., Gallium Arsenide, in which P-region is doped with Germanium and n-region with Tellurium. The thickness of the P-N junction layer is very narrow so that the emitted laser radiation has large divergence. The junctions of the ‘p’ and ‘n’ are well polished and are parallel to each other as shown in fig. (a)

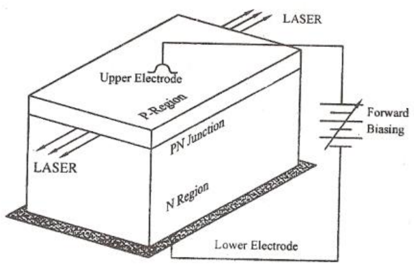


Fig.(a) Semiconductor Laser

Since the refractive index of GaAs is high, it acts as optical resonator; the external mirrors are not needed. The upper and lower electrodes fixed in the 'p' and 'n' region helps for the flow of current to the diode while biasing.

Working:

1. The population inversion in a p-n junction is achieved by heavily doping 'p' and 'n' materials. So that the Fermi level lies within the conduction band of n type and within the valence band of 'p' type as shown in fig.(b)

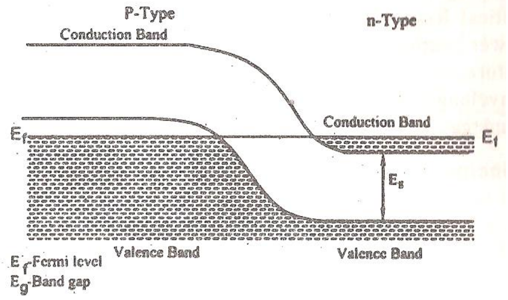


Fig. (b)

2. If the junction is forward biased with an applied voltage nearly equal to the band gap voltage, direct conduction takes place. Due to high current density active region is generated near the depletion region.
3. At this junction, if a radiation having frequency (ν) is made to incident on the p-n junction then the photon emission is produced as shown in fig.(c)
4. Further, the emitted photons increases the rate of recombination of injected electrons from the n region and holes in p region by inducing more re - combinations.

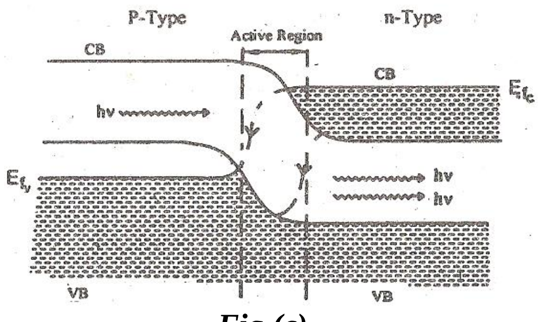


Fig (c)

5. Hence the emitted photons have the same phase and frequency as that of original inducing photons and will be amplified to get intense beam of laser.
6. The wavelength of the emitted radiation depends on
   (i) The band gap and (ii) The concentration of donor and acceptor atoms in GaAs.

Advantages

1. It is easy to manufacture the diode.
2. The cost is low.

Disadvantages

1. It produces low power output.
2. The output wave is pulsed and will be continuous only for some time.
3. The beam has large divergence

2. Explain the construction and working of a Nd-YAG laser (April 2011, May 2013, 14 Jan 2015,Nov 2019)

Characteristics of Nd- YAG Laser

1. Type: Doped insulator laser (solid state laser)
2. Active medium: Yttrium Aluminium garnet ($Y_3Al_5O_{12}$)
3. Active medium: Neodymium (Nd^{3+} ions)
4. Pumping method: Optical pumping
5. Pumping Source: Xenon flash lamp
6. Optical Resonator: Ends of the rods polished and silvered and two mirrors one of which is totally reflecting and the other partially reflecting.
7. Power output: 2 x 10^4 Watts
8. Wavelength emitted: 1.064 µm.

Introduction:

Nd-YAG laser is a doped insulator laser. It is a four level system in which the active medium is taken in the form of a crystal. Here the crystal is intentionally doped during its growth. Those types of lasers have number of energy levels with same energy. The laser is used to generate high power intensity.

Principle:

The term "Doped Insulator Laser" refers to the active medium, Yttrium Aluminium Garnet doped with Neodymium Nd^{3+} ions has many energy levels. Due to optical pumping these ions are raised to excited levels. During the transition from metastable state to E_1 state, the laser beam of wavelength 1.064 µm.

Construction:

The active medium is made as a rod which has Yttrium aluminium garnet ($Y_3Al_5O_{12}$) doped with a rare earth metal ion Neodymium Nd^{3+} ions . The Nd^{3+} ion normally occupies the Yttrium ions and provides the energy levels for both lasing transitions and pumping. This rod is placed inside a highly reflecting elliptical cavity as shown in fig(a)

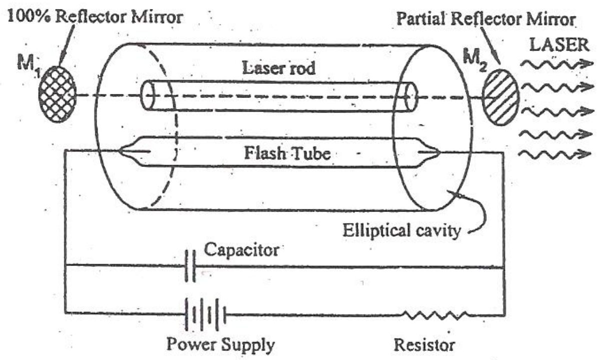


Fig. (a) Nd – YAG Laser

A close optical coupling is made by placing the Xenon flash lamp near by the laser rod, in such a way that most of the radiations from the flash tube passes through the laser rod due to the elliptical cavity. The flash tube may be switched ON and controlled with the help of a capacitor. The discharge of capacitor is initiated using a high voltage source.

The optical resonator is formed by grinding the ends of the rods and coated with silver accompanied by two mirrors, one is 100% reflecting and the other is partially reflecting which, is included to increase the efficiency of the output beam.

Working:

1. The xenon flash lamp is switched ON and the light is allowed to fall on the laser rod.
2. The intense white light excites the Neodymium (Nd^{3+}) ions from the ground state to various energy levels above E_2. Hence the atoms are raised to group of higher levels in E_3 as shown in Fig. (b).

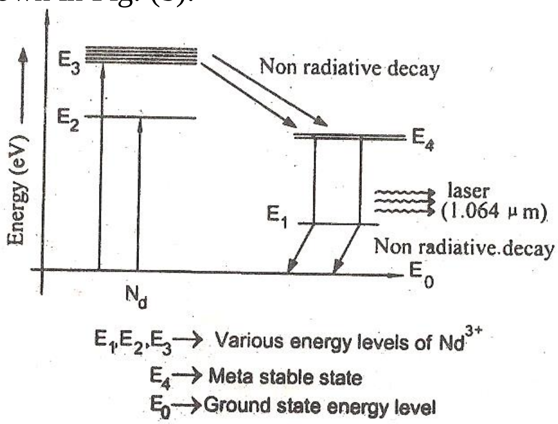


Fig. (b) Energy level of Nd-YAG Laser

3. From these energy levels the ions make non- radioactive decay and are gathered in a state called as Metastable state, until the population inversion is achieved.
4. Once the population inversion is achieved, the stimulated is achieved.
5. Hence pulses of laser beam of wavelength 1.064 µm is emitted during the transition from E_4 to E_1 (lower)
6. A large amount of heat is produced by the flash tube during the working. Hence cooling arrangement is made either by blowing air (or) circulating water over the crystal.

Applications of Nd-YAG Laser:

1. It is used in transmitting signals to longer distances.
2. It is used in long haul communication system.
3. It is also used in the endoscopic applications.
4. It plays a vital role in remote sensing applications.

3. Discuss the construction and working of CO_2 laser (Jan 2011, 2014, May 2012,15,16 ,17,18,Nov 2017,19)

Characteristics of CO_2 laser

1. Type: Gas Laser
2. Active medium: Mixture of CO_2, N_2 and Helium (or) Water vapour.
3. Active centre: CO_2
4. Pumping method: Electric discharge method
5. Optical Resonator: Metallic mirror of gold (or) silicon mirrors coated with Aluminium.
6. Power output: 10 KW
7. Nature of output: continuous (or) pulsed

8. Wavelength of the output: 9.6 µm & 10.6 µm

Principle:

The transition between these vibrational and rotational energy levels leads to the construction of molecular gas laser. Here the Nitrogen atoms are initially raised to excited state. The nitrogen atoms deliver the energy to CO_2 atoms which has closet energy level to it. Then, transition takes place between the vibrational energy levels of the CO_2 atoms and hence laser beam is emitted.

The molecular gas lasers have two types of transitions.

(1) Transitions between vibrational states of the same electronic state (Fig. a.).

(2) Transitions between vibration levels of different electronic state (Fig. b)

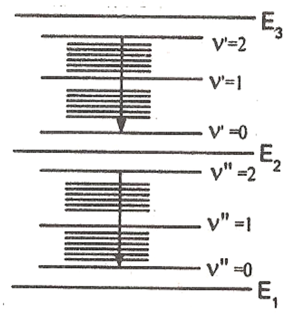


Fig. a

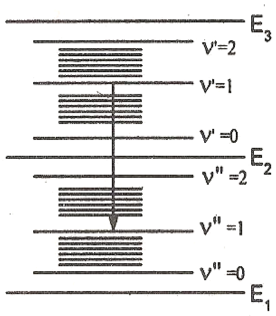


Fig. b

CO_2 laser satisfies the first condition, here the laser transition occurs between vibrational energy levels of the same electronic state.

Fundamental modes of vibration of CO_2 molecule

(i) **Symmetric stretching mode($10^{o}0$)**

Here the carbon atom is stationary and the oxygen atoms oscillate or vibrate along the axis of the molecule as shown in fig. (c)

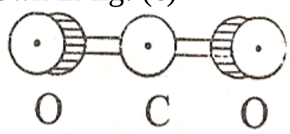


Fig. (c) Symmetric mode

The two oxygen atom simultaneously approaches or departs with respect to carbon atom. The state of vibration is given by 3 integers (m n l q) here ($10^{o}\theta$) which corresponds to the degree of excitation.

(ii) **Asymmetric stretching mode ($00^{o}1$), ($00^{o}2$)**

Here all the three atoms will vibrate. Here the oxygen atoms vibrate in the opposite direction to the vibration of carbon atom as shown in fig (d)

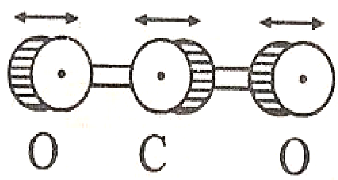


Fig. (d) Asymmetric mode

It gives the quanta of frequency ($00^{o}1$), ($00^{o}2$)

(iii) **Bending mode ($01^{o}0$, $02^{o}0$)**

Here the atoms will not be linear rather the atoms will vibrate perpendicular to the molecular axis as shown in fig. (e) This gives rise to two quanta of frequency represented by ($01^{o}0$, $02^{o}0$)

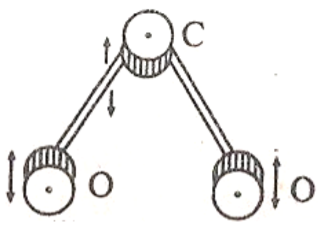


Fig (e) Bending mode

Construction:

It consists of a discharge tube in which CO_2 is taken along with nitrogen and helium gases with their pressure of 0.33:1.2:7mm of Hg for CO_2, nitrogen and the He respectively. Nitrogen helps to increase the population of atoms in the upper level of CO_2. While Helium helps to depopulate the atoms in the lower level of CO_2 and also to cool the discharge tube. The discharge is produced by D.C excitation. At the ends of the tube sodium chloride/Brewster windows are placed as shown in fig (f) Confocal silicon mirrors coated with aluminum or metallic mirror of gold is employed for proper reflection, which form the resonant cavity. The output power can be increased by increasing the diameter of the tube.

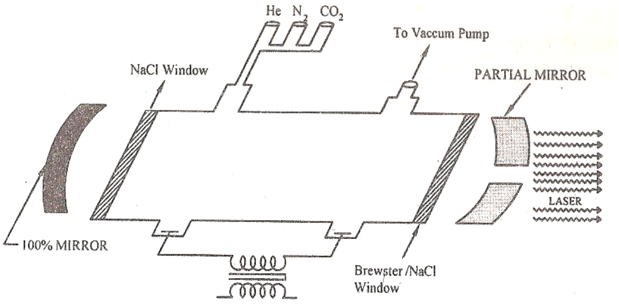


Fig. (f) CO_2 Laser

Working:

1. The discharge is passed through the tube first the nitrogen atoms are raised to excited state.

$$e + N_2 \longrightarrow N_2^*$$

2. The excited N_2 atoms undergo resonant energy transfer with CO_2 atom and raises CO_2 **(**00º1) to excited state due to closer energy level CO_2 **(**00º1) and Nitrogen

$$N_2^* + CO_2 \longrightarrow CO_2^* + N_2$$

3. When transition takes place between 00º1 to 10º0, laser of wavelength 10.6μm is emitted as shown in fig (e)
4. Similarly, when transition takes place between 00º1 and 02º0 laser beam of wavelength 9.6 μm is emitted as shown below.
5. Since 00º1 $\rightarrow$ 10º0 has a higher gain than 00º1 $\rightarrow$ 02º0 transitions. Usually the laser beam of wavelength 10.6 μm is produced more.
6. When the gas flow is longitudinal power output is 50 to 60 watts but if the gas flow is perpendicular to the discharge tube the output power may be raised to 10 Kw/m.

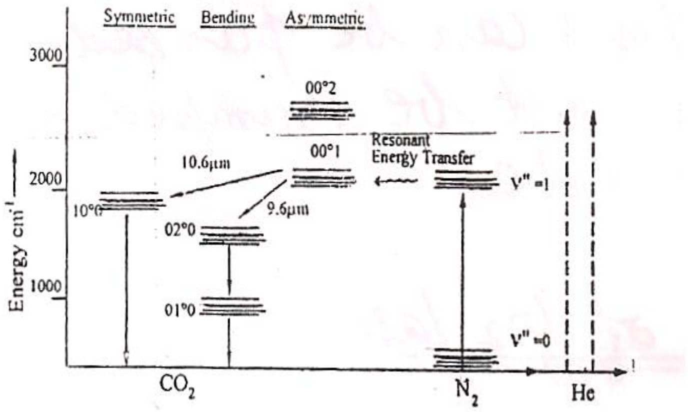


Fig.(e) Energy Level Diagram

7. This type of CO_2 laser is known as TEA laser (Transversely Excited Atmospheric Pressure Laser)
8. The combination of carbon monoxide and oxygen will also have some effect on the laser action. To avoid these unused gases can be pumped out and fresh CO_2 must be pumped inside the discharge tube.

Application of CO_2 Laser

1. This laser has applications in medical field such as neurosurgery, treatment of liver, lungs and also is bloodless operations.
2. It is widely used in open air communications.
3. This layer also has wide application over military field.

4. Using Einstein's theory of stimulated absorption, spontaneous emission and stimulated emission. Obtain the relationship between Einstein coefficients A and B. (April 2011, Jan 2016,May 2014,16,Sept 2020)

Determination of Einstein's coefficients (A & B)

We know that, when light is absorbed by atoms (or) molecules, then they are excited from the lower energy level (E_1) to the higher energy level (E_2) and during the transition from higher energy level (E_2) to know energy level (E_1) the light is emitted from the atoms (or) molecules.

According to Einstein it is impossible to predict which atom is going to be transition from lower energy level (E_1) to the higher energy level (E_2). But it is possible to calculate the rate of transition by using the law of probability.

When an atom is exposed to (light) photons of energy E_2-E_1=hυ, three distinct process take place.

(i) Absorption
(ii) Spontaneous emission
(iii) Stimulated emission

(i) Absorption:

An atom in the lower energy level or ground state energy level E_1 absorbs the incident photon radiation of energy hυ and goes to the higher energy level (or) excited energy state E_2 as shown in fig (a) This process is called as **absorption.**

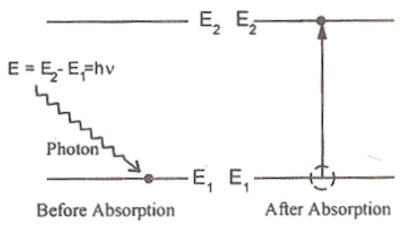


Fig (a) Absorption

If there are many number of atoms in the ground state then each atom will absorb the energy from the incident photon and goes to the excited state then, the rate of absorption R_{12} is proportional to the following factors

i.e. R_{12} α energy density of incident radiation (ρ_υ)

R_{12} α No : of atoms in the ground state N_1

i.e. $R_{12} \alpha \rho_\upsilon N_1$ ……………… (1)

(or) $R_{12} = B_{12}\rho_\upsilon N_1$

Where B_{12} is a constant which gives the probability of absorption transition per unit time. Normally, the atom in the excited state will not stay there for a long time, rather it comes to ground state by emitting a photon of energy E=hυ such an emission takes place by one of the following two methods.

(ii) Spontaneous emission:

The atom in the excited state returns to the ground state by emitting a photon of energy **E= E$_2$-E$_1$=hυ,** spontaneously without any external triggering as shown in fig (b). This process is known as **spontaneous emission**.

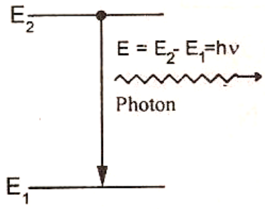


Fig (b) Spontaneous emission

Such an emission is random and is independent of incident radiation. If N_1 and N_2 are the no. of atoms in the ground state (E_1) and excited state (E_2) respectively, then the rate of spontaneous emission is

$R_{21}(SP) \alpha N_2$

$R_{21} (SP) = A_{21}N_2$ ……………….. (2)

(iii) Simulated emission:

The atom in the excited state can also return to the ground state by external triggering (or) inducement of photon thereby emitting a photon of energy equal to the energy of the incident photon, known as **stimulated emission**. Thus results in two photons of same energy, phase difference and of same directionality as shown in fig. (c)

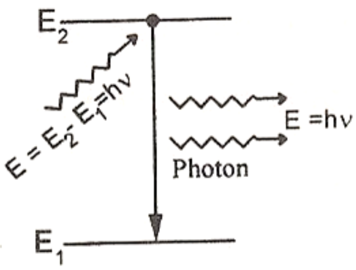


Fig. (c) Stimulated Emission

The rate of stimulated emission is given by

$$R_{21} \alpha \rho_\upsilon N_2$$

$$R_{21} = B_{21}\rho_\upsilon N_2 \qquad \text{..... (3)}$$

Where B_{21} is a constant, which give the probability of stimulated emission transition per unit time.

Expressions for the coefficients A_{21}, B_{12} and B_{21}

Einstein's theory of absorption and emission of light by an atom is based on Planck's theory of radiation. Also under thermal equilibrium, the population of energy levels obey the Boltzmann's distribution law.

i.e. Under thermal equilibrium

The rate of absorption = The rate of emission

i.e. Eq(1) = Eq(2) + Eq(3)

$$B_{12}\rho_\upsilon N_1 = A_{21}N_2 + B_{21}\rho_\upsilon N_2 \qquad \text{..... (1)}$$

$$\rho_\upsilon (B_{12}N_1 - B_{21}N_2) = A_{21}N_2 \qquad \text{..... (2)}$$

$$\rho_\nu = \frac{A_{21} - N_2}{B_{12}N_1 - B_{21}N_2} \text{ ..... (3)}$$

Dividing the above equation both on Numerator and Denominator by N_2

$$\rho_\nu = \frac{A_{21}}{B_{12}\left(\frac{N_1}{N_2}\right) - B_{21}} \text{ ..... (4)}$$

We know form Boltzmann's distribution law

$$N_1 = N_o e^{(-E_1/K_B T)}$$

Similarly $N_2 = N_o e^{(-E_2/K_B T)}$

Where, K_B is the Boltzmann's constant

T is the absolute temperature and

N_o is the number of atoms at absolute zero.

At equilibrium, we can write the ratio of population levels as follows.

i.e.

$$\frac{N_1}{N_2} = e^{(E_2 - E_1)/K_B T}$$

Since $E_2 - E_1 = h\upsilon$, we have

$$\frac{N_1}{N_2} = e^{h\nu/K_B T} \text{ .... (5)}$$

Substituting eqn (5) in eqn (4) we have

$$\rho_\nu = \frac{A_{21}}{B_{12}(e^{h\nu/K_B T}) - B_{21}}$$

Divide the above equation both on Numerator and Denominator by B_{21}

$$\rho_\nu = \frac{A_{21}}{B_{21}} \frac{1}{\left[\frac{B_{12}}{B_{21}} e^{(h\nu/K_B T)} - 1\right]} \text{ ....(6)}$$

This equation has very good agreement with Planck's energy distribution radiation law.

$$\rho_v = \frac{8\pi h v^3}{c^3} \frac{1}{e^{(h\nu/K_B T)} - 1} \ldots\ldots\ldots\ldots(7)$$

Therefore comparing eqn (6) and eqn (7) we can write $B_{12} = B_{21} = B$ …..(8)
and

$$\frac{A_{21}}{B_{21}} = \frac{8\pi h v^3}{c^3} \ldots\ldots\ldots\ldots\ldots (9)$$

Taking $A_{21} = A$ and the constants A and B are called as **Einstein Coefficients.**

RESULT:

(i) Equation (8) states that the stimulated emission rate per atom is the same as the absorption rate per atom.

(ii) Equation (9) gives the relation between the spontaneous emission and stimulated emission coefficients. Since the ratio is proportional to γ^3, the probability of spontaneous emission rapidly increases with the energy differences between the two states.

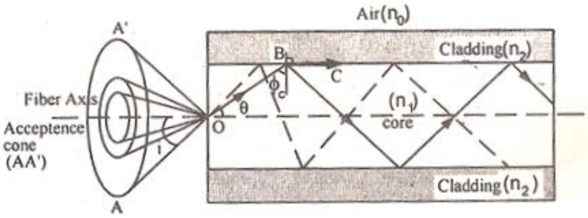
5. What is meant by acceptance angle for an optical fibre? Show how it is related to numerical aperture? (May 2012, 2013, Jan 2016)

(OR)

Derive an expression for the acceptance angle and numerical aperture of an optical fibre. (Jan 2013,14,15, May 2014,16, Sept 2020)

Let us consider a cylindrical fibre. It consists of core of refractive index n_1 and cladding of refractive index 'n_2' and let 'n_0'be the refractive index of the medium (air) in which the optical fibre is placed.

The incident ray travel along AO and enters the core at an angle 'i'. The ray is refracted along OB at an angle θ in the core as shown in fig



Fig

It further proceeds to fall at critical angle of incidence ϕ_c =90-θ on the interface between core and cladding. At this angle the ray just moves along BC. Any ray which enters into the core at an angle of incidence less than i will have refractive angle less than θ.

Hence the angle of incidence (ϕ=90-θ) at the interface of core and cladding will be more than the critical angle. Hence the ray is totally internally reflected ray. Thus, only those ray which passes within the acceptance angle (cone) will be totally reflected (internally). Therefore, the light incident on the core within this maximum external incident angle (i_m) can be coupled into the fibre to propagate. This angle (i_m) is called as wave **acceptance angle**.

Expression for Acceptance angle and Numerical aperture:

(i) Applying Snell's law at point of entry of ray (AO) we have

$n_0 \sin i = n_1 \sin\theta$

$\sin i = \frac{n_1}{n_0}$

$\sin i = \frac{n_1}{n_0} (\sqrt{1 - cos^2 \theta})$ ……… (1)

(ii) Applying Snell's law at point B (I.e. on interface)

$n_1 \sin\phi = n_2 \sin 90^o$ ……… (2)

$\sin\phi = \frac{n_2}{n_1}$ (since sin 90º = 1)

$$\text{Sin } (90-\theta) = \frac{n_2}{n_1}$$

$\cos \theta = \frac{n_2}{n_1}$ ……… (3)

substituting eqn (3) in eqn (1) we get

$$\text{Sin } i = \frac{n_1}{n_0}\sqrt{1 - \frac{n_2 2}{n_1 2}}$$

$$\text{Sin } i = \frac{n_1}{n_0} \cdot \frac{1}{n_1}\sqrt{{n_1}^2 - {n_2}^2}$$

$$Sin\ i = \frac{\sqrt{n_1^2 - n_2^2}}{n_0}$$

$$i = \sin^{-1} \frac{\sqrt{n_1^2 - n_2^2}}{n_0}$$

The angle $i = \sin^{-1} \frac{\sqrt{n_1^2 - n_2^2}}{n_0}$ is the acceptance angle.

If the medium surrounding the fibre is air $n_0=1$

$$i = \sin^{-1} \sqrt{n_1^2 - n_2^2}$$

Numerical Aperture (NA) :

It is defined as sine of acceptance angle.

$$NA = \sin i$$

$$NA = \sqrt{n_1^2 - n_2^2}$$

Acceptance angle:

The maximum angle at or below which the light can suffer **total internal reflection is called as acceptance angle**. The cone is referred as acceptance cone.

6. What is population inversion? Explain why laser action cannot occur without population inversion between atomic levels?
(May 2011)

The state of achieving more number of atoms in the excited state **(N_2)** than the ground state **(N_1)** is called as **population inversion.$N_2 > N_1$**.

Consider two energy level systems E_2 and E_1 representing upper and lower state .We know that when a photon of energy equal to the differences in energy between two levels approaches the two atom, both the stimulated emission and absorption process are equally possible.

Under thermal equilibrium the population of the energy levels obey Boltzmann's distribution given by the equation.

$$\mathbf{N_1 = \exp [-(E_2 - E_1)/kT]}$$

Let N_1 be the number of atoms in ground state

N_2 be the number of atoms in excited state. Then, For stimulated emission to dominate, it is necessary to increases the population of the upper energy level so that this is greater than lower level. This is known as **“population inversion”.** The material having a population inversion is called **“active material”**

Condition for population inversion:

(i) There must be at least a pair of energy levels.

(ii) There must be a source to supply energy to the medium.

(iii) The atoms must be raised continuously to the excite state.

7. Explain (May 2011, 2017 Nov 2017)

(i) Step index fibre and

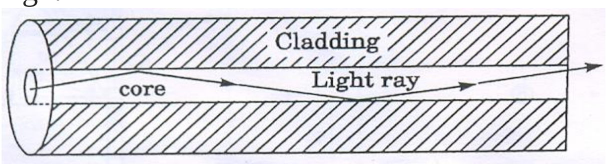
1. Step-index fibre:

In step index fibre, the variation in refractive indices of air, cladding and core vary step by step. Hence this type of fibre is known as **STEP-INDEX** fibre.Based on the refractive index and the number of modes step index fibre is classified into

- Step index- single mode fibre.
- Step index – multi mode fibre.

Step index- single mode fibre:

1. It consists of a thin core of uniform refractive index of a higher value
2. The core is surrounded by a cladding of uniform refractive index lesser than that of the core.
3. Core diameter is 5 to 10µm external diameter of cladding is 50 to 125µm.
4. Due to its small core, only a single mode of light ray transmission is possible .
5. Numerical aperture is small.
6. Loss of light is less.
7. Band width is high.



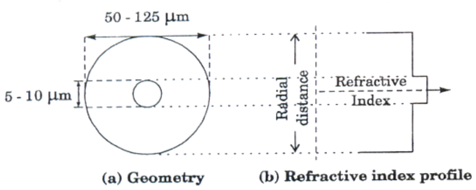
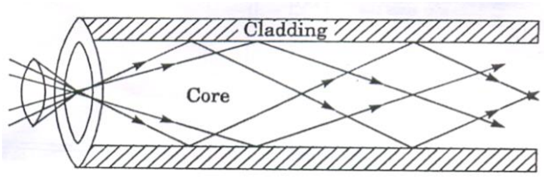


Fig. (a & b) Single mode step up index fibre

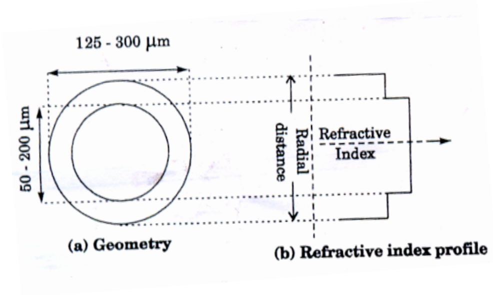
Step index – multi mode fibre.

1. Its core has much larger diameter which makes it easier to support propagation of a larger number of modes.
2. It has a core material with uniform refractive index and cladding material of lesser refractive index than that of the core.
3. A typical step-index multimode fibre has a core diameter of 50 to 200µm and external diameter of cladding 125µm to 300µm.
4. Because of larger diameter of the core, the propagation of many modes the fibre is possible.
5. Numerical aperture is high.

6. Loss is large.
7. Bandwidth is less



(a)



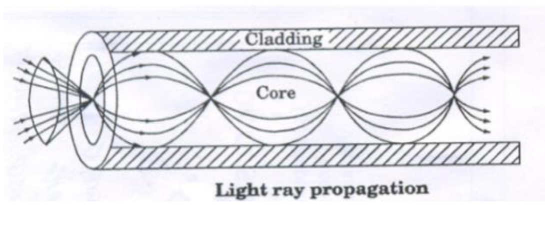
(b)

Fig. (a&b) Multimode Step up Index Fibre

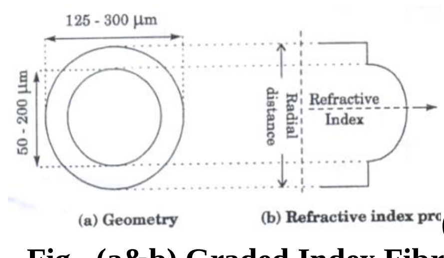
(ii) Graded index fibre

Graded-index fibre:

1. The refractive index of the core is maximum along the fibre axis and it gradually decreases towards core-cladding interface. Thus it is called as graded index fibre.
2. A typical graded index multimode fibre has a core diameter of 50 to 200µm, and an external diameter of cladding 100 to 250µm.
3. If the core is high, the intermodal dispersion loss must be high.
4. The refractive index of the core is uniform throughout and undergoes step change at the cladding boundary.
5. Attenuation is more for multimode Step Index Fibre but less for single mode Step Index Fibre.
6. Numerical Aperture is more for multimode Step Index Fibre but less for single mode Step Index Fibre.
7. Step index multi mode fibre has a lower band width



(a)



(b)

Fig. (a&b) Graded Index Fibre

8. Write an essay on the preparation, advantages of optical fibre communication (May 2015,16 Jan 2014,Nov 2016,2017)

A fibre optic communication system is very much similar to a traditional communication system. Its purpose is to transfer information from a source to a distant user.

Principle:

The basic principle of optic fibre communication is transmission of information over a required distance by the propagation of optical signal through optical fibre.

Description:

The main parts of fibre optical communication system are,

1. Information signal source
2. Transmitter
3. Light source
4. Propagation medium
5. Receiver

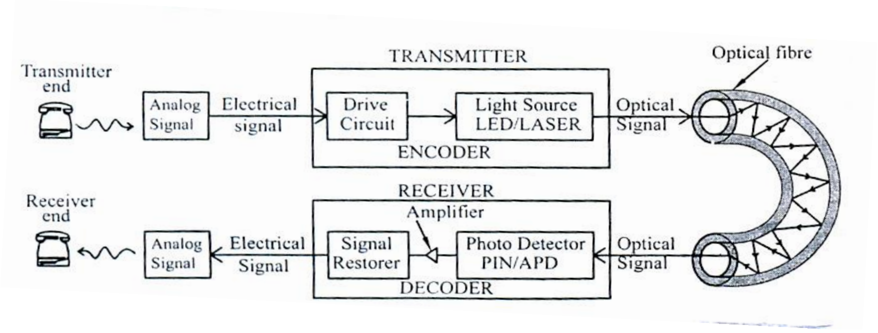


Fig. Schematic diagram of Fibre optic communication system

1. Information signal source:

The information signal source may be voice, music, digital data or analog voltage and video signal. Here it is analog information.

2. Transmitter:

It consists of a drive circuit and a light source. The drive circuit transfers the electric input signals (information signal) into digital pulses.

3. Light source:

The light source generates optical energy which acts as the information carrier. Laser or LED is used for light sources. This light source converts digital pulses into optical pulses.

4. Propagation medium

The optical fibre is used as a propagation medium. It acts as a waveguide and transmits optical pulses towards receiver by the principle of total internal reflection.

5. Receiver:

It consists of a photo detector, an amplifier and a signal – restoring circuit. The photo detector detects optical signal and converts back into an electrical signal. The signal is amplified and it converts from digital to analog signals.

Working:

Analog information such as the voice of a telephone user produces electrical signals in analog form this analog electrical signal is converted into digital signal which is in the form of electrical pulses. The electrical signal modulated the light emitted by an optical source (such as a LED or laser diode). Now, this optical signal is fed into the fibre.

At the receiving end, the optical signal passing through the fibre is fed into a photo detector. The photo detector detects optical signal and converts it into pulses of electric current. This digital signal is once again converted into analog signal. This analog signal contains the same information as transferred from transmitting end. By this way, the information is transmitted from one end to other end.

9. Describe the propagation of light through an optical fibre (May 2017, Sept 2020)

For optical fibres, the process of propagation of light is simple, because once the light enters the fibre; the rays do not encounter any new surfaces but repeatedly they hit the same surface. The reason of confining the light beam inside the fibre is the **total internal reflection**.

Even for a bent fibre, the light guidance takes place by multiple total internal reflections all over the length of the fibre as shown in fig (a)

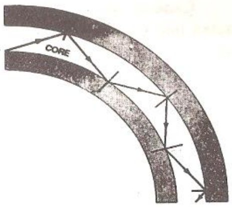


Fig (a)

Principle: The principle of optical fibre communication is total internal reflection.

Total Internal Reflection:

The phenomenon of total internal reflection takes place when it satisfies the following two conditions.

Condition 1: Light should travel from denser medium to rarer medium

i.e. $n_1>n_2$

Where n_1= refractive index of core

n_2= refractive index of cladding

Condition 2: The angle of incidence on core should be greater than the critical angle

i.e., $\phi>\phi_c$

Where ϕ- angle of incidence

ϕ_c- Critical angle

Propagation phenomenon:

Let the light ray traverses from denser medium to rarer medium.

Case (i): When $\phi<\phi_c$, the ray traverses along the interface as shown in below figure (b)

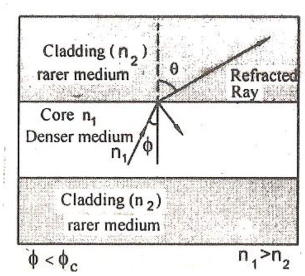


Fig (b)

Case (ii): When $\phi=\phi_c$, the ray traverses along the interface as shown in below Fig. (c)

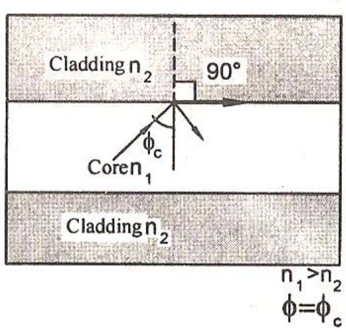


Fig (c)

So, that the angle of refraction is 90º. This angle ϕ_c is called as **critical angle**.
Case(iii) :when $\phi > \phi_c$, the ray is totally reflected back (internally) into the denser medium itself as shown in the below figure (d)

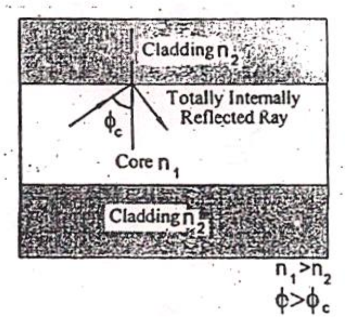


Fig (d)

From Snell's law (The maximum angle for total internal reflection ϕ_c)

$$n_1 \sin \phi_c = n_2 \sin 90^o$$

$$\sin\phi_c = \frac{n_2}{n_1}$$

$$\phi_c = sin^{-1} \left(\frac{n_2}{n_1}\right)$$

10.(a) Write a note on temperature sensor. (May 2014)

Principle:

The absorption of light by the silicon layer varies with temperature and the variation modulates the intensity of the light received at the detector

Description:

It consists of a light source, a detector, two fibres (transmitting fibre, receiving fibre) and silicon covered with a reflecting coating as shown in fig.

Working:

1. When the light is passed through the fibre from the source and the light after reflecting from the silicon layer returns to a detector. During this process, light is absorbed by the silicon layer.

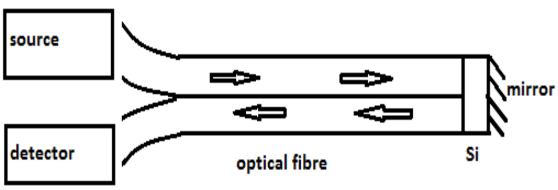


Fig. Temperature Sensor

2. The absorption of light by silicon layer varies with temperature and this variation modulates the intensity of the light received at the detector.
3. A photo detector is used to measure the intensity of light and temperature is sensed. Temperature in the range from 80^oC to 700^0C is measured using this technique.

(b) State the application of fibre optic sensor.(May 2016)

1. It is used as liquid level indicator based on the principle of change in refractive index of the medium.
2. Optical displacement sensors are used to find the position and displacement of the target.
3. Temperature sensors are used to measure the temperature accurately.
4. The pressure can accurately measured using photo elastic pressure sensors.
5. Laser Doppler velocimeter (LDV) is used to measure several physical quantities such as velocity = fluid surface velocity.